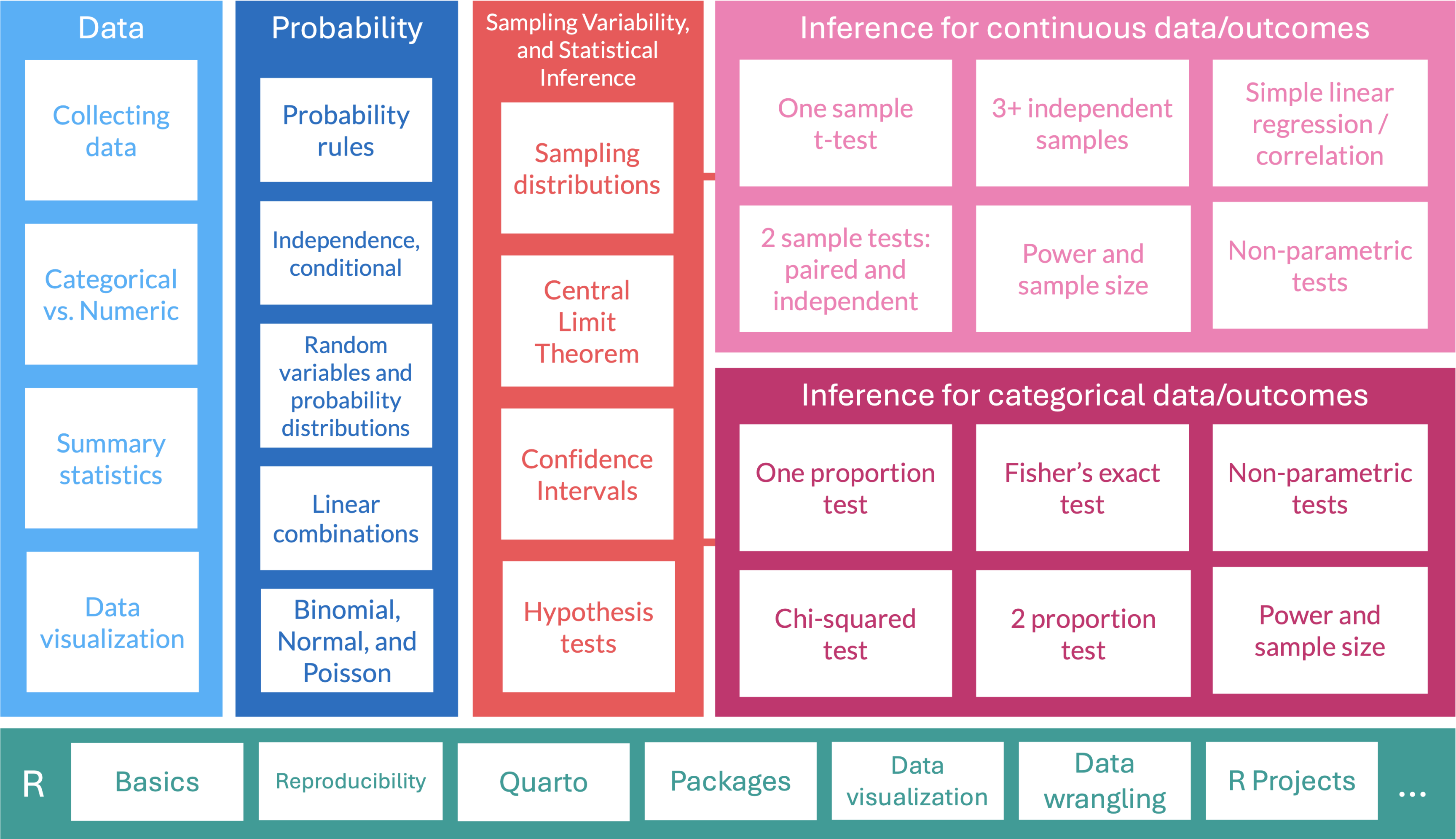


# R04: Reproducibility, File Organization, and Projects

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# Where are we?



# Using OneDrive

- We all have free access to OneDrive to store files
- **Let's login** into our online accounts
- **You can also download OneDrive for your desktop**
  - Allows you to access the OneDrive from your computer's interface instead of the browser
  - Creates a link between your computer and the cloud!
- Let me show you mine
  - I can access all the files through RStudio as well!
- Let's take a couple minutes to log into OneDrive

# Folder organization

- Make a folder for our class!
  - I suggest naming it something like `EPI_525_F24` to indicate the class and the term
- Make these folders in your computer
  - Only make them in OneDrive **if** you have a desktop connection
- For a project, I have the following folders
  - Background
  - Code
  - Data\_Raw
  - Data\_Processed
  - Dissemination
  - Reports
  - Meetings
- For our class, I suggest making one folder for the course with the following folders in it:
  - Data
  - Homework
  - Lessons
  - R\_activities
  - And other folders if you want
- Take a few minutes to create these folders



## Aside: folder and file naming

There are a few good practices for naming files and folders for easy tracking:

1. Keep the name short and relevant
2. Use leading numbers to help organize sequential items
  - I can show you my lessons folders as an example
3. Use dates in the format “YYYY-MM-DD” so that files are in chronological order
4. You can label different versions if you would like to
5. Use “\_” to separate sections of the name
  - I also use this to separate words, but some people say you should use “-” to separate words

# Let's go into R\_activities

- Make a folder called something like R04\_work
- We are going to stay in the R\_activities folder
- Now that we have some folders in order, we can start creating a project

# Creating project in RStudio

- Way to designate a working directory: basically your home base when working in R
  - We have to tell R exactly where we are in our folders and where to find other things
  - A project makes it easier to tell R where we are
- Basic steps to create a project
  - Go into RStudio
  - Create new project for this class (under **File** or top right corner)
    - I would chose “Existing Directory” since we have already set up our folders
    - Make the new project in the **EPI\_525\_F24** folder
- Once we have projects, we can open one and R will automatically know that its location is the start of our working directory

# The nice thing about R projects

- 5 minute video explaining some of the nice features of R projects

<https://rfortherestofus.com/2022/10/rstudio-projects>



# Reproducibility

- Research data and code can reach the same results regardless of who is running the code
  - This can also refer to future or past you!
- We want to set up our work so the entire folder can be moved around and work in its new location

# Nicky's process for R projects

- Let me show you my process
  - I will create one in my Sample\_folder
  - I will show you how I switch between classes
- Make a project file in your class folder!